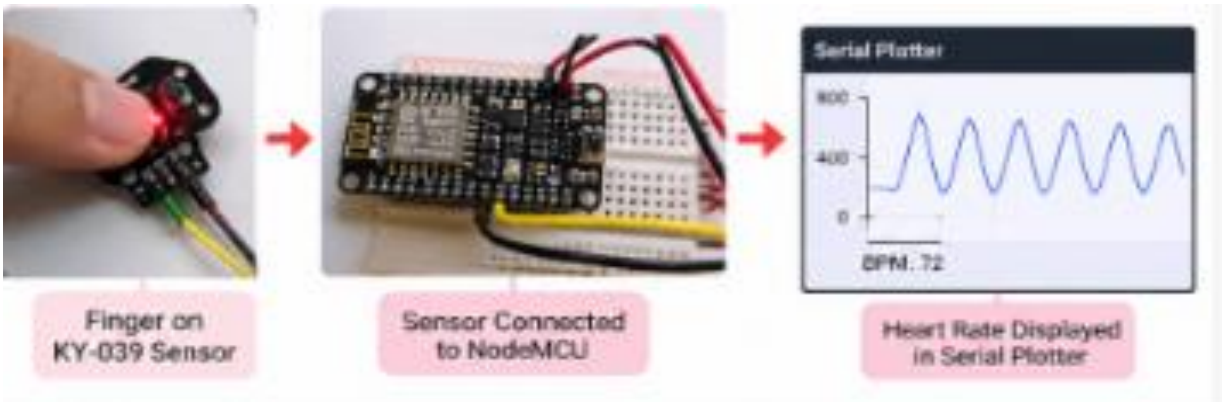
Experiment - 9

Batch - 8

Aim:- To find and display the heart rate (BPM – Beats Per Minute) using the KY-039 heartbeat sensor and ESP8266 NodeMCU in the Serial Plotter.

Equipment Required:-

1. KY-039 Heartbeat Sensor
2. ESP8266 NodeMCU
3. Connecting wires



Working:-

The KY-039 sensor detects changes in blood flow using infrared light.

1. The IR LED emits infrared light.
2. Place your finger on the sensor.
3. Blood flow changes whenever the heart beats.
4. The phototransistor detects changes in light intensity.
5. The sensor sends an analog signal to the ESP8266 through A0.
6. The program detects heartbeat peaks and calculates BPM.
7. The heart rate is displayed in the Serial Plotter.

Connections:-

KY-039	NodeMCU
S	A0
+ / VCC	5V
− / GND	GND

Code:-

```
#define PULSE_PIN A0
unsigned long lastBeatTime = 0;
unsigned long currentTime;
int rawValue;
int previousValue = 0;
int BPM = 0;
int threshold = 15;
bool beatDetected = false;
void setup()
{    Serial.begin(115200);
   pinMode(PULSE_PIN, INPUT);
   Serial.println();
   Serial.println("KY-039 Heartbeat Sensor");
   Serial.println("Place your finger gently on the sensor");
   Serial.println("----------------------------------------");}
```

```
void loop()
{   rawValue = analogRead(PULSE_PIN);
     currentTime = millis();
    if ((rawValue- previousValue) > threshold &&  !beatDetected)
    {   unsigned long interval = currentTime- lastBeatTime;
        if (interval > 300 && interval < 2000)
        {   BPM = 60000 / interval;
            Serial.print("Heartbeat detected BPM = ");
            Serial.println(BPM);}
        lastBeatTime = currentTime;
        beatDetected = true;}
    if (rawValue < previousValue)
    {    beatDetected = false;}
    previousValue = rawValue;
    Serial.print("Signal:");
    Serial.print(rawValue);
    Serial.print("\tBPM:");
    Serial.println(BPM);
    delay(10);
}
```

Outputs:-

